

■ ONION QUALITY DETECTION & GRADING SYSTEM

Smart India Hackathon (SIH Challenge 26031) | Technical Specification & Documentation

Status: Production Ready | Architecture: Multi-Platform (FastAPI + YOLOv8 + React + Capacitor)

1. Executive Summary & Problem Statement

Manual inspection across agricultural procurement centers is inherently **subjective, time-intensive, and prone to human error**. Inspectors evaluating truckloads of onions lack standardized digital verification tools, creating discrepancies between farmers and procurement agencies.

This project introduces an automated AI grading workstation that captures onion imagery, segments individual bulbs, calculates physical and morphological metrics, detects internal and external defects, and compiles instant tamper-proof inspection certificates.

2. System Capabilities & AI Workflow

Three-Phase Processing Architecture:

- **Phase 1: Image Acquisition:** Supports multi-angle capture from webcams, industrial cameras, or mobile devices with dynamic lighting compensation and multi-onion batch segmentation.
- **Phase 2: Computer Vision & Defect Analysis:** Employs YOLOv8 instance segmentation (yolov8n-seg.pt) for contour polygon extraction, diameter/area estimation, color distribution analysis, and anomaly identification (sprouting, root growth, fungal rot, mechanical cuts).
- **Phase 3: Grading & Certification:** Classifies each item into standardized commercial tiers (Grade A, Grade B, Grade C, or Reject) and renders cryptographic-ready inspection reports with human-in-the-loop override records.

Quality Grading Standards

Grade	Criteria & Specifications	Primary Market Destination
Grade A	Diameter > 55mm, uniform shape, zero rot/sprouting, intact skin (< 5% blemish)	Export & Premium Retail
Grade B	Diameter 40-55mm, minor surface marks (< 15%), intact shape, no soft rot	Domestic Supermarkets & Wholesale
Grade C	Diameter 25-40mm, irregular shape, superficial discoloration, safe for processing	Commercial Dehydration / Culinary
Reject	Active sprouting, soft neck rot, black mold spores, severe mechanical crush	Unfit for Human Consumption

3. Technical Architecture & Tech Stack

Component	Technology	Responsibilities
AI / ML Engine	PyTorch, Ultralytics YOLOv8, OpenCV, NumPy	Instance segmentation, polygon masking, colorimetric evaluation, defect scoring
Backend API	FastAPI, Python 3.10+, Uvicorn, SQLite	REST API endpoints, batch queueing, database audit trails, ReportLab PDF rendering
Frontend Web	React 18, TypeScript, Vite, Tailwind/Vanilla CSS	Interactive inspector dashboard, real-time image capture, override justification modal
Mobile App	Capacitor, Android Native Shell, Gradle	Offline-ready field inspection APK for handheld tablets and smartphones

4. Core REST API Endpoints

Method	Endpoint	Description	Payload / Response
GET	/health	Health check & model status	Status JSON
POST	/api/analyze	Executes segmentation & grading on batch images	Multipart Image → Detections JSON
POST	/api/override-grade	Records human judge override with reason	BatchID, OnionID, NewGrade, Justification
POST	/api/generate-report	Generates formal quality certificate	BatchID → Downloadable PDF
GET	/api/batches	Retrieves historical batch inspections	Array of Batch Summaries

5. Execution & Deployment Guide

A. Starting the Backend API:

Open terminal in project root and execute run-backend.bat or:

```
cd sih_onion
.\env_deploy\Scripts\activate
uvicorn backend.main:app --host 0.0.0.0 --port 8000 --reload
```

B. Launching Web Dashboard:

Open terminal in project root and execute deploy-website.bat or:

```
cd moa_app
npm install
npm run dev
```

C. Compiling Android Mobile APK:

Run deploy-android.bat or build via Capacitor CLI:

```
cd moa_app
npm run build && npx cap sync && npx cap open android
```

In Android Studio, select **Build > Build Bundle(s) / APK(s) > Build APK(s)**.